SOFTWARE TEST STRATEGY AND TEST PLAN

IEM Admission Portal

Team 404 Brain Not Found | PCCCS594 — Software Engineering and Project Management Laboratory | Semester V

Prepared using the project SRS, project charter and HLD worksheet. The sample admission test plan was used only as a structural reference.

Project and team details

Student / Team Detail	Information
Project title	IEM Admission Portal
Team name	404 Brain Not Found
Team / Group	10
Course	PCCCS594 — Software Engineering and Project Management Laboratory
Section / Batch	C / 2024–2028
Semester / Year	5th Semester / 3rd Year
Faculty in-charge	Sangeeta Sarkar; Sadhu Prasad Kar
Team members	Debarghya Mitra (248) — Project Manager; Arka Ghosh Dastider (220) — UI/UX Designer; Jitesh Jana (243) — Quality Assurance (QA) Engineer; Jatin Kumar (232) — Software Architect; Diptomoy Das (227) — Full-Stack Engineer

PART A — SOFTWARE TEST STRATEGY DOCUMENT

A.1 Introduction and Objectives

This document defines the overall approach for verifying and validating the IEM Admission Portal, a web-based platform intended to simplify student admissions. Applicants can register, complete and submit applications, upload required documents, pay application fees and track application status. Authorized administrators can manage submitted applications, verify documents and update application status.

The objective of testing is to check that the implemented system satisfies the approved Software Requirements Specification (SRS), works correctly across the applicant and administrator workflows, protects applicant records from unauthorized access, and is usable for the intended users. Testing will identify defects before deployment and support the project acceptance criteria: implementation of approved requirements, no critical defects, successful UAT and approval by the project supervisor.

A.2 Scope

The strategy covers the modules and workflows identified in the project SRS and HLD worksheet:

Module / Subsystem	Testing coverage
Applicant registration and management	Registration using required personal and contact information; validation and handling of incomplete input.
Application form and submission	Completing, saving/submitting and managing an admission application.
Document upload and verification	Applicant document upload and administrator document-verification workflow.
Application fee payment	Fee submission and handling of payment status; integration dependency on the payment service.
Application status tracking	Display of the current status to the applicant and status updates by authorized administrators.
Administrative management	Authorized access to application records, application review, document verification and status updates.
End-to-end admission workflow	Registration → application completion/submission → document upload → fee payment → administrative review/status update → applicant tracking.

The exact admission eligibility thresholds, document types/file-size limits, payment-provider details and status-transition rules are not specified in the supplied SRS. They must be confirmed by the project stakeholders before corresponding detailed acceptance tests are finalized.

A.3 Levels of Testing

A.3.1 Unit Testing

Individual functions and components will be tested in isolation by the developers. Examples include applicant registration validation, application-form validation, document-upload handling, payment-status handling, application-status retrieval and administrator authorization checks. Test inputs will include valid, missing, malformed and boundary data where limits are defined. Developers will correct defects and rerun the affected unit tests.

A.3.2 Integration Testing

An incremental, top-down integration approach will be used for the main admission workflow. The applicant-facing workflow will first be connected to controlled stubs for services that are not yet available, such as the fee-payment service and document-verification component. The stubs will then be replaced by the actual components when available. Integration testing will check data exchange, status consistency, error handling and access control between the applicant portal, application management, document verification, payment service and administrator portal.

A.3.3 System Testing

The fully integrated portal will be tested as a complete system against the SRS. Functional tests will cover registration, application submission, document upload, fee payment, status tracking and administrator operations. Non-functional tests will cover the stated performance target (normal user requests within 3 seconds, excluding external-service delays), security of applicant records and usability for applicants with basic computer and web-application knowledge.

A.3.4 Acceptance Testing

User Acceptance Testing (UAT) will be performed by representative applicants and authorized admission/administrative users, with review by the project supervisor. Alpha-style evaluation may be conducted in the project test environment with the team and institutional representatives. A limited trial with selected users may be used as beta-style feedback if permitted. Acceptance will be based on the approved requirements, successful completion of the admission workflow, absence of critical defects and final supervisor approval.

A.4 Testing Techniques Used for Test-Case Design

A.4.1 Black-box techniques

• Equivalence Class Partitioning — divide inputs into valid and invalid groups. Apply to registration fields and application forms: complete required information versus one or more missing required fields; authorized versus unauthorized user access.

• Boundary Value Analysis — test values at, just below and just above specified limits. Apply to the SRS performance requirement of 3 seconds for normal user requests (excluding external-service delays). Other field-length, file-size or file-type boundaries will be added only after their limits are specified.

• Decision Table Testing — check combinations of application completion, document submission/verification and payment status to verify that the portal displays and records the appropriate workflow state. The precise business rules and permitted status transitions must be confirmed by stakeholders.

• State Transition Testing — test application status changes initiated through the administrator workflow and verify that the applicant sees the current status. Only transitions approved by the admissions process should be treated as valid.

A.4.2 White-box techniques

• Statement Coverage — execute every executable statement in selected validation, status-management and authorization functions at least once.

• Branch / Decision Coverage — exercise both true and false outcomes for conditions such as required-field validation, authorization checks and payment-status handling.

• Basis Path Testing — use the control-flow graph and cyclomatic complexity of a selected application-readiness validation function to identify independent paths. A worked representative example is provided in Part B. The implementation code should be substituted if the final code differs.

A.5 Testing Types

• Functional testing — registration, application completion/submission, document upload, fee payment, status tracking, document verification and administrator status updates.

• Regression testing — rerun relevant existing tests after code changes or defect fixes, especially when applicant data validation, payment handling, document processing or status management is changed.

• Performance/load testing — verify the SRS target of responding to normal user requests within 3 seconds under the expected project workload, excluding external-service delays; include concurrent applicant activity where the test environment permits.

• Security testing — verify that applicant information and application records are accessible only to authorized users; test attempts by one applicant to view or modify another applicant's records and attempts to access administrator functions without authorization.

• Usability testing — evaluate whether applicants with basic computer and web-application knowledge can register, complete the application, upload documents, pay the fee and track status with minimal assistance.

A.6 Debugging Approach

When a test fails, the QA engineer will record the steps, test data, expected result and actual result. The team will use log/database inspection (brute-force examination) to locate errors, backtracking from the incorrect screen or status to the responsible function, and cause elimination to isolate the failing validation, integration or authorization condition. The responsible developer will fix the defect. The QA engineer will rerun the failed test and related regression tests, and the test lead will verify closure.

A.7 Entry and Exit Criteria

Testing Level	Entry Criteria	Exit Criteria
Unit Testing	Function/component implemented; build succeeds; unit test cases and expected results prepared.	All planned unit tests executed; no known critical unit defect remains; relevant statement/branch coverage achieved.
Integration Testing	Connected components have passed unit testing; required stubs/drivers and test data are available.	Planned interfaces and data exchanges tested; no open critical integration defect; status and error handling behave as expected.
System Testing	All required modules integrated; test environment configured; representative applicant/admin test accounts and data available.	Approved SRS requirements tested; performance, security and usability checks completed; no open severity-1/critical defect.
Acceptance Testing	System testing completed; UAT environment and scenarios ready; representative users and supervisor available.	UAT scenarios accepted; critical defects closed; project supervisor approval recorded.

A.8 Roles and Responsibilities

Role / Assigned member	Responsibility
Project Manager — Debarghya Mitra (248)	Coordinate the test schedule and resources, track progress and risks, and communicate test status to the team and supervisor.
UI/UX Designer— Arka Ghosh Dastider (220)	Design and maintain user interfaces, ensure ease of navigation, and verify interface consistency and usability.
QA Engineer — Jitesh Jana (243)	Design test cases, execute functional/integration/system/regression tests, record results and report defects.
Software Architect — Jatin Kumar (232)	Review test coverage against system architecture and interfaces; support analysis of integration and design-related defects.
Full-Stack Engineer — Diptomoy Das (227)	Implement and unit-test application components, investigate defects and provide fixes for retesting.
Applicant / Administrator representative	Participate in UAT and confirm that applicant and administrative workflows meet the agreed needs.
Project supervisor	Review acceptance evidence and provide final project approval.

A.9 Test Tools and Environment

The test environment will contain the web application, its required application/data services, test applicant and administrator accounts, and representative non-production records. Test data should be fictional or anonymized. The project documents do not mandate specific testing products; the team may use an issue tracker such as JIRA or Bugzilla for defect logging, a unit-testing framework appropriate to the chosen implementation language, browser-based manual testing and automated API/UI tests where available. Payment and document-verification dependencies may be represented by stubs until their real integrations are available.

A.10 Risk-Based Considerations

• Unauthorized access or exposure of applicant records — prioritize role/access-control tests because the SRS requires applicant information and application records to be available only to authorized users.

• Payment-service or document-verification integration failure — prioritize integration and error-handling tests because the project depends on external services for fee payment and may depend on document-verification components.

• High request volume or slow response during admission activity — prioritize performance testing against the SRS target of 3 seconds for normal requests, excluding external-service delays.

• Requirement ambiguity — confirm eligibility criteria, required document rules, payment outcomes and permitted status transitions with stakeholders before treating them as fixed acceptance rules.

PART B — SOFTWARE TEST PLAN

Test plan focus: Applicant Registration, Application Submission, Document Upload, Fee Payment and Application Status Tracking

B.1 Test Plan Identifier

TP-IEMAP-ADMISSION-01

B.2 Objective

To verify that the IEM Admission Portal supports the applicant admission workflow described in the SRS: an applicant can register, complete and submit an application, upload required documents, submit the application fee and track the current application status; authorized administrators can manage applications, verify documents and update status.

B.3 Features to be Tested

• Applicant registration using required personal and contact information.

• Completion and submission of the admission application.

• Upload of required documents and handling of document-verification status.

• Application-fee submission and recording/handling of payment status.

• Applicant access to the current status of their own submitted application.

• Administrator access to manage applications, verify documents and update application status.

• End-to-end interaction between applicant, application, document, payment and administrative components.

B.4 Features Not to be Tested (Out of Scope)

• Counselling, seat allocation and admission-merit/rank calculation — these functions are not specified in the supplied IEM Admission System SRS.

• Generation/formatting of admit cards, hall tickets or offer-letter PDFs — not specified in the supplied SRS/HLD worksheet.

• Production payment-provider certification or real-money transactions — testing will use a test/sandbox service or stub unless the project explicitly provides an approved test integration.

B.5 Business Rule(s) / Feature Under Test

The portal must support the following documented workflow: an applicant registers with required personal and contact information, completes and submits an admission application, uploads the required documents and submits the application fee. The applicant must be able to view the current status of the submitted application. An authorized administrator must be able to manage submitted applications, verify documents and update application status. Applicant information and application records must be accessible only to authorized users.

The supplied SRS does not define numerical eligibility cut-offs, exact mandatory document types, file-size limits, payment amount, or a complete list of valid status transitions. Tests in this plan therefore verify the documented functions and access rules without inventing those policy values. Stakeholder-approved rules should be added when available.

B.6 Black-Box Test Design — Equivalence Partitioning & Boundary Value Analysis

Input under test: Application response time for a normal user request. The SRS specifies a response time of within 3 seconds under the expected project workload, excluding delays caused by external services.

Class Type	Description	Representative value(s)
Valid equivalence class	Normal request completes within the stated target.	2.5 seconds
Boundary — valid side	Request completes just below the 3-second target.	2.9 seconds
Boundary — exact target	Request completes at the stated 3-second target.	3.0 seconds
Boundary — invalid side	Request takes just longer than the stated target.	3.1 seconds
Invalid equivalence class	Request takes clearly longer than the stated target.	5.0 seconds

Performance test cases

TC ID	Description	Input (response time)	Expected Output	Actual Output	Pass / Fail
TC-B01	Normal request within target	2.5 s	Request completes within the 3-second target.	To be recorded	Pending
TC-B02	Just below target	2.9 s	Request completes within the 3-second target.	To be recorded	Pending
TC-B03	At target boundary	3.0 s	Request meets the stated 3-second target.	To be recorded	Pending
TC-B04	Just above target	3.1 s	Performance target is not met; record as a defect/observation.	To be recorded	Pending
TC-B05	Clearly above target	5.0 s	Performance target is not met; record as a defect/observation.	To be recorded	Pending

Note: Measure application response time without including delays caused by external services, as specified by the SRS. The examples are test input durations, not measured results.

B.7 Black-Box Test Design — Decision Table Testing

Conditions: C1 = Required application information completed?; C2 = Required documents uploaded?; C3 = Application fee status is paid? Action: application workflow readiness/status handling. Since the SRS does not explicitly define a submission-gating policy, the expected results below check that the system records/displays the relevant condition and does not falsely report missing steps as completed.

Rule	C1: Form complete	C2: Documents uploaded	C3: Fee paid	Expected handling
R1	Yes	Yes	Yes	Record the completed workflow information; application can proceed to administrative review according to the configured process.
R2	Yes	Yes	No	Show/retain payment as pending or unpaid; do not show fee payment as completed.
R3	Yes	No	Yes	Show document upload as incomplete; do not show required documents as uploaded.
R4	Yes	No	No	Show both document and payment steps as incomplete/pending.
R5	No	Yes	Yes	Identify incomplete application information; do not represent the application form as complete.
R6	No	Yes	No	Show incomplete application information and unpaid fee status.
R7	No	No	Yes	Show incomplete application information and missing document status.
R8	No	No	No	Show the application information, document and fee steps as incomplete/pending.

Representative decision-table test cases

TC ID	Input Set 1: Form	Input Set 2: Documents	Input Set 3: Fee	Expected Result
TC-D01 (R1)	Complete	Uploaded	Paid	Portal records the steps and displays the current application status for the applicant.
TC-D02 (R2)	Complete	Uploaded	Unpaid	Portal indicates payment is not completed; it must not display a paid status.
TC-D03 (R3)	Complete	Not uploaded	Paid	Portal indicates required document upload is incomplete.
TC-D04 (R5)	Incomplete	Uploaded	Paid	Portal identifies missing application information; form is not represented as complete.
TC-D05 (R8)	Incomplete	Not uploaded	Unpaid	Portal indicates the relevant workflow steps are incomplete.

These cases validate consistent status representation. The final rule for whether an application can be submitted or reviewed with a pending step must be confirmed by the admissions stakeholders.

B.8 White-Box Test Design — Basis Path Testing

Module under test: validateApplicationReadiness(formComplete, documentsUploaded, feePaid) — a representative helper that checks whether the three documented workflow steps are complete. This pseudocode is a test-design example, not a claim about the team's current implementation.

Numbered representative code:

```
1  boolean validateApplicationReadiness(boolean formComplete,
2                                       boolean documentsUploaded,
3                                       boolean feePaid) {
4      if (!formComplete) {
5          return false;
6      }
7      if (!documentsUploaded) {
8          return false;
9      }
10     if (!feePaid) {
11         return false;
12     }
13     return true;
14 }
```

Control-flow graph: Nodes = {1, 4, 5, 7, 8, 10, 11, 13}; Edges = {1→4, 4→5, 4→7, 7→8, 7→10, 10→11, 10→13}. Decision nodes = 3. Cyclomatic complexity V(G) = number of decision nodes + 1 = 3 + 1 = 4. Therefore, 4 linearly independent paths are required for basis-path coverage.

Path #	Path description	Input values (form, docs, fee)	Expected result
P1	Form incomplete; first condition returns false.	(false, true, true)	false — readiness validation fails.
P2	Form complete; documents missing; second condition returns false.	(true, false, true)	false — readiness validation fails.
P3	Form and documents complete; fee unpaid; third condition returns false.	(true, true, false)	false — readiness validation fails.
P4	All three conditions are true; function reaches final return.	(true, true, true)	true — all represented workflow steps are complete.

The four paths exercise each decision's true and false outcomes and cover all independent paths in this representative function. If the implemented function differs, update the node/edge list and recalculate its cyclomatic complexity.

B.9 Integration Test Cases

Integration strategy: Incremental top-down. Start with the applicant/application workflow and use stubs for the payment service and document-verification component where required. Replace each stub with the real component when it becomes available.

TC ID	Integration step	Description	Expected result
TC-I01	Step 1 — controlled stubs	Applicant/application component communicates with payment and document-verification stubs.	Application workflow receives the stubbed document and payment statuses and displays them consistently.
TC-I02	Step 2 — payment integration	Replace the payment stub with the available payment-service test integration; test paid and unpaid outcomes.	The portal records/displays the returned payment status accurately and handles unsuccessful/pending outcomes without marking them paid.
TC-I03	Step 3 — document integration	Connect the document-upload workflow to the document-verification component.	Uploaded documents and verification results are associated with the correct applicant/application.
TC-I04	Step 4 — administrator integration	Connect application management and administrator status-update functions.	Authorized administrator updates are reflected in the applicant's current status view.
TC-I05	Full integration	Run registration → application submission → document upload → fee payment → administrator review/status update → applicant tracking.	The end-to-end workflow completes and the applicant sees the current recorded status; records remain associated with the correct applicant.

B.10 System Test Cases (End-to-End, Black-Box)

TC ID	Scenario	Steps	Expected result
TC-S01	Applicant registration	Open registration; enter required personal/contact information; submit.	Applicant registration is processed and the user receives the appropriate system response.
TC-S02	Incomplete registration/application data	Leave a required field empty and attempt to submit.	System identifies the missing required information and does not represent the data as complete.
TC-S03	Application submission	Register; complete the application form; submit the application.	Application submission is processed and the applicant can view its current status.
TC-S04	Document upload	Open the document-upload area and upload the required documents.	System accepts/processes the upload according to configured document rules and associates it with the correct application.
TC-S05	Fee payment	Initiate fee payment using the approved test/sandbox flow.	System records/displays the payment result returned by the payment service; unpaid/failed payment is not shown as paid.
TC-S06	Applicant status tracking	Sign in as an applicant and open the submitted application's status view.	Applicant can view the current status of their own application.
TC-S07	Administrator document verification	Sign in as an authorized administrator; open an application; review its submitted documents and update the verification information.	Authorized administrator can manage the application and the recorded document status is updated.
TC-S08	Administrator status update	Sign in as an authorized administrator; update an application's status.	System saves the update and the applicant's status view reflects the current recorded status.
TC-S09	Unauthorized record access	Sign in as Applicant A and attempt to view or modify Applicant B's application.	Access is denied; Applicant A cannot view or modify Applicant B's record.
TC-S10	Normal-request performance	Execute representative normal user requests under expected project workload and measure response time, excluding external-service delays.	Normal user requests complete within the SRS target of 3 seconds.

B.11 Test Deliverables

• Software Test Strategy and Test Plan document.

• Test-case specifications for black-box, white-box, integration and system testing.

• Test execution/results log showing actual result and pass/fail status.

• Defect reports containing steps to reproduce, severity, evidence and resolution status.

• Regression test results after defect fixes.

• UAT feedback and acceptance/sign-off record.

• Final test summary report.

B.12 Suspension and Resumption Criteria

Testing will be suspended if a severity-1/critical defect prevents further execution, such as a system crash, loss/corruption of application records, incorrect association of one applicant's data with another applicant, or a security failure that exposes applicant records. Testing may also be paused if a required integration or test environment is unavailable and prevents meaningful execution. Testing will resume after the defect or environment issue is fixed, the corrected build is deployed to the test environment, and the QA engineer verifies the fix with the failed test and relevant regression tests.

B.13 Approval

Name	Role	Signature	Date
Jitesh Jana	QA Engineer / Test Plan Prepared By	_____	_____
Debarghya Mitra	Project Manager	_____	_____
Project Supervisor	Client / Project Representative	_____	_____

Approval confirms that the test approach and scope are accepted. Any unresolved policy details identified in this document must be confirmed before final UAT test cases are baselined.